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Key Points:

- Parallel factor analysis components of all leaf litter leachates shifted from more protein-like to more humic-like with senescence
- Phenophase and species had a greater influence on the fluorescent dissolved organic matter than the geographic origin of beech trees
- Behavior of the tyrosine-like component differed significantly between two beech genetic clusters, suggesting differences in resorption

Supporting Information:

- Supporting Information S1

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Tracking senescence-induced patterns in leaf litter leachate using parallel factor analysis (PARAFAC) modeling and self-organizing maps

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Abstract In autumn, the dissolved organic matter (DOM) contribution of leaf litter leachate to streams in forested watersheds changes as trees undergo resorption, senescence, and leaf abscission. Despite its biogeochemical importance, little work has investigated how leaf litter leachate DOM changes throughout autumn and how any changes might differ interspecifically and intraspecifically. Since climate change is expected to cause vegetation migration, it is necessary to learn how changes in forest composition could affect DOM inputs via leaf litter leachate. We examined changes in leaf litter leachate fluorescent DOM (FDOM) from American beech (*Fagus grandifolia* Ehrh.) leaves in Maryland, Rhode Island, Vermont, and North Carolina and from yellow poplar (*Liriodendron tulipifera* L.) leaves from Maryland. FDOM in leachate samples was characterized by excitation-emission matrices (EEMs). A six-component parallel factor analysis (PARAFAC) model was created to identify components that accounted for the majority of the variation in the data set. Self-organizing maps (SOM) compared the PARAFAC component proportions of leachate samples. Phenophase and species exerted much stronger influence on the determination of a sample's SOM placement than geographic origin. As expected, FDOM from all trees transitioned from more protein-like components to more humic-like components with senescence. Percent greenness of sampled leaves and the proportion of tyrosine-like component 1 were found to be significantly different between the two genetic beech clusters, suggesting differences in photosynthesis and resorption. Our results highlight the need to account for interspecific and intraspecific variations in leaf litter leachate FDOM throughout autumn when examining the influence of allochthonous inputs to streams.

1. Introduction

Climate change impacts forest ecosystems by spawning temporal shifts in phenology and vegetation migration [*Parmesan and Yohe*, 2003]. Biogeographic shifts in species composition are prompted by increases in air temperatures and alterations in precipitation patterns. These shifts have been forecasted to affect the range of entire species [*Iverson and Prasad*, 2002] and shown to affect the intraspecific genetic variation of certain tree species with latitude [*O'Neill et al.*, 2008]. But, how such shifts in the biogeography of tree species affect the biogeochemical cycling of forests remains inadequately understood. It is, however, widely accepted that higher temperatures are lengthening the growing season in boreal and temperate forests as (on average) the timing of budburst advances and the timing of senescence retreats [*Richardson et al.*, 2010]. Autumn phenology, in particular, is more complicated to predict and understand than spring phenology [*Gallinat et al.*, 2015]. Autumn phenological events are primarily controlled by temperature and photoperiod, but genetics and water stress also play major roles [*Killingbeck*, 2004]. Disentangling the complexities of autumnal phenology is important to achieve a more comprehensive understanding of forest biogeochemical cycling.

The dissolved organic matter (DOM) in a forested stream is highly dependent on its end-members, including leaf litter leachate, precipitation, throughfall, groundwater, and stemflow. Particularly in the autumn, leaf litter leachate plays a significant role in forest nutrient cycles. Leaves can contribute organic matter to a stream either by direct submergence or by temporary submergence in runoff. Upon submergence, up to 40% of the initial leaf mass can be leached within the first several days [*Boulton*, 1991; *McDowell and Fisher*, 1976]. McDowell and Fisher [1976] also demonstrated that autumnal DOM utilization accounted for nearly 10% of annual ecosystem respiration and 77% of that originated from leaf litter leachate inputs. Additional material can be leached from leaves during freeze-thaw cycles throughout the winter, which can be transported to nearby streams via snowmelt and runoff [*Bärlocher*, 2005]. *Inamdar et al.* [2013] found that among their sampled end-members, leaf litter leachate had the highest temporal variation. Leaf litter is responsible for 29% of the 99% of allochthonous annual energy input into forested, headwater streams [*Fisher and Likens*, 1973].

In the autumn, nutrient concentrations in leaves dramatically change as physiological processes, such as resorption, senescence, and abscission occur [*Killingbeck*, 1996]. Nitrogen (N) resorption is expected to be altered with climate change, which would impact the N concentrations in abscised leaves [*Estiarte and Peñuelas*, 2015]. While fallen leaves do contribute N to streams, nitrate concentrations have been shown to decrease significantly in the autumn with leaf fall due to increased carbon availability for in-stream heterotrophic uptake of N [*Goodale et al*., 2009], but changes in N resorption will likely affect this process. Leaf litter leachate studies have mostly focused on postsenesced leaves that were abscised off the trees without a disturbance. However, disturbances, such as hurricanes, can cause leaf fall (i.e., greenfall) throughout the year, and thus, understanding the DOM in leaf litter leachate at different phenological stages is important. Additionally, depending on when disturbances occur, the nutrient concentrations and thus the chemistry of the resultant leaf litter leachate should differ. But how it will differ is still poorly understood.

Characteristics of the chromophoric fraction of DOM (CDOM) in leaf litter leachate can be used to quantify the bioavailability of DOM. CDOM can be measured using ultraviolet-visible (UV-Vis) absorbance spectroscopy or fluorescence excitation-emission matrix (EEMs) spectroscopy [Coble, 1996; *Cory et al*., 2011]. Parallel factor analysis (PARAFAC) modeling has been used extensively in the literature to analyze the variation in EEMs data sets [*Cory and McKnight*, 2005; *Stedmon and Bro*, 2008]. While used in various other disciplines, Kohonen's self-organizing maps (SOMs) are a new, recent technique to further analyze EEMs data by helping to characterize and identify patterns within a EEMs data set [*Cuss and Guéguen*, 2016]. Similar to Tobler's First Law of Geography, samples that map closer together on the SOM are more similar and those that map further apart are more different. SOMs are a method of mapping high-dimensional data onto a two-dimensional grid of nodes [*Kohonen*, 2001] and can be used to identify similar and less similar samples and patterns in the samples [*Cuss et al*., 2016]. DOM samples can be mapped on SOMs using the whole processed EEMs, excitation/emission pairs, PARAFAC component loadings, and PARAFAC component proportions. SOMs of whole EEMs can show more variation between samples than just the component loadings and proportions from PARAFAC modeling but do not allow for identification and characterization of the DOM fluorophores [*Cuss et al*., 2014]. SOMs likely have the potential to increase the understanding of and the application of EEMs and PARAFAC data.

Several studies have either included leaf litter leachates in their PARAFAC model data set [*Singh et al*., 2013; *Cuss and Guéguen*, 2016] or have made leaf litter leachate specific data sets [*Beggs and Summers*, 2011; *Cuss et al*., 2014]. *Beggs and Summers* [2011] analyzed lodgepole pine needles collected in August, and *Cuss and Guéguen* [2016] and *Cuss and Guéguen* [2013] analyzed senescent and overwintered leaf litter leachate from several species including sugar maple, white spruce, and silver maple. These studies, however, are not capable of showing how the role of tree leaves in a watershed's DOM dynamics changes throughout autumn, especially because autumn phenology is expected to shift by a magnitude of days [*Jeong et al*., 2011]. A better understanding of the effects of the seasons, and their transitions, on leaf leachate is especially critical given DOM impacts on nutrient dynamics and microbial activity in both soils and streams in a watershed [*Bernhardt and Likens*, 2002; *Fellman et al*., 2009; *Wymore et al*., 2015; *Joly et al*., 2016]. To the best of our knowledge, no leaf litter leachate PARAFAC models have been published that include leaves spanning three different phenophases: green, senescing, and recently abscised. Furthermore, while PARAFAC models have been created using different tree species, we are not aware of any that specifically include members of the same species, but from different genetic populations.

For this study, we were interested in the three following primary research questions: (1) how do the PARAFAC components of leaf litter leachate change throughout the autumn?, (2) how might the PARAFAC components of leaf litter leachate in a forest be affected by community restructuring?, and (3) how useful is the new technique of SOMs at visualizing and analyzing these potential differences? We created a PARAFAC model and SOMs from a leaf litter leachate data set that includes American beech (*Fagus grandifolia* Ehrh.) and yellow poplar (*Liriodendron tulipifera* L.) leaf litter leachates in three phenophases to help visualize autumnal change in leaf litter leachate DOM. Additionally, our data set includes leaf litter leachate samples from beech trees in the two distinct genetic clusters and four different states in the eastern United States (Vermont, Rhode Island, Maryland, and North Carolina) to elucidate how the change in leaf litter leachate FDOM differs between these different ecosystems for a single species and whether differences are dependent on the genetic clusters.

2. Materials and Methods

2.1. Site Descriptions

Leaves collected in Maryland (MD) were gathered at Fair Hill Natural Resource Management Area in Cecil County (39.71°N, 75.85°W). This site receives 1205 mm mean annual precipitation. The 1981–2010 30 year mean maximum air temperature is 19.1°C, and the 30 year mean minimum temperature is 6.8°C [*National Climatic Data Center (NCDC)*, 2016]. Soils in the area are primarily loam and silt loams of igneous, metamorphic, phyllite, and schist parent material, specifically Baile silt loam with 0–3% slopes, Delanco-Codorus-Hatboro complex with 0–8% slopes, Glenelg loam with 8–15% slopes, and Manor loam with 15–25% slopes [*U.S. Department of Agriculture Natural Resources Conservation Service (USDA NRCS)*, 2012] at 70 m elevation. Soils were found to be 0.129% total N. The forest type is mixed deciduous, containing primarily yellow poplar, beech, and sweet birch (*Betula lenta*).

Rhode Island (RI) leaves were collected in an old-growth beech forest found within the Aquidneck Land Trust's Oakland Forest in Newport County (41.56°N, 71.26°W). The 1981–2010 30 year mean maximum temperature is 15.1°C, and the 30 year mean minimum temperature is 6.4°C [*NCDC*, 2016]. The average annual precipitation is 1176 mm. Soils in the site consist of Newport, Pittstown, and Stissing silt loams, with slopes of less than 8% [*USDA NRCS*, 2012] at an elevation of about 47 m and consist of 1.065% total N.

In Vermont (VT), leaves were collected in the Wade Brook watershed near Montgomery Center (44.86°N, 72.55°W) near an elevation of 390 m. Primary species in this forest are sugar maple, red maple, yellow birch, American beech, and white ash. The 1981–2010 30 year mean maximum temperature is 13.3°C, and the 30 year mean minimum temperature is 1.1°C [*NCDC*, 2016]. The average annual precipitation is 1060 mm. Soils in the site are primarily Peru fine sandy loam, with 3 to 15% slopes, at times very stony [*USDA NRCS*, 2012], and are 0.355% total N.

Leaves from North Carolina (NC) were collected on Rich Mountain, Boone, NC, above 1400 m elevation (36.24°N, 81.71°W). This site receives 1337 mm annual mean precipitation. The 1981–2010 30 year mean maximum temperature is 16.3°C, and the 30 year mean minimum temperature is 4.0°C [*NCDC*, 2016]. Soils in the site consist of Burton-Craggey-Rock Outcrop with 15 to 95% slopes (gravelly to cobbly sandy loam profile) and Burton-Wayah Complex with 15 to 50% slopes (gravelly loam to fine sand-loam to gravelly loamy sand profile) [*USDA NRCS*, 2012]. Soils consist of 0.382% total N.

All soil samples were collected in August 2016 and analyzed by the University of Delaware's Soil Testing Lab for total N and other elements (interested readers are referred to Table S1 in the supporting information for additional elements). This lab participates in the North American Proficiency Testing Program for Agricultural Testing Laboratories operated by the Soil Science Society of America, the USDA Northeastern Coordinating Committee for Soil Testing NEC1312, and the Mid Atlantic Soil Testing and Plant Analysis Workgroup.

2.2. Leaf Collection and Leachate Preparation

Thirty beech leaves were collected at each site for three phenophases (green, senescing, and freshly abscised), totaling 360 leaves. Leaves were collected at each site from multiple trees of various ages and diameters to avoid an age-induced confounding influence. All green and senescing leaves that were collected at the four sites were accessible from the ground and do not include any upper canopy leaves. Field sites in VT, RI, and MD were selected due to their proximity to the different collaborating universities, while American beech was chosen as the primary species in our study due to its pervasiveness throughout the eastern United States and presence at the three initial field sites. The field site in western NC was included to add another American beech population that has been found to be genetically similar to that in VT [*Kitamura and Kawano*, 2001]. MD beech leaves were collected on 29 June 2015, 5 November 2015, and 11 November 2015 for green, senescing, and fallen phenophases, respectively. RI beech leaves were collected on 6 July 2015, 6 November 2015, and 15 December 2015. VT beech leaves were collected on 13 July 2015, 19 October 2015, and 30 October 2015. NC beech leaves were collected on 25 August 2015, 12 October 2015, and 25 October 2015. Based on *Kitamura and Kawano* [2001], MD and RI beech leaves were classified as belonging to the mild-environment genetic cluster and VT and NC as the harsh-environment cluster. Additionally, a total of 90 yellow poplar leaves were collected at the MD site for the same phenophases on 21 July 2015, 26 October 2015, and 1 November 2015. Yellow poplar leaves were only collected from the MD site given its prevalence at the study site and to serve as a comparator to beech.

Leaves from RI, VT, and NC were shipped to the University of Delaware, and all leaves were stored for 48 h between collection and measurement to minimize potential differences caused by differing transit times. All leaves were photographed, and light reflectances were measured on all senescing and abscised leaves using a PSP-1100 Field-Portable Spectrophotometer (Spectral Evolution Inc., Lawrence, Massachusetts). All leaves were briefly rinsed with Nanopure® deionized water, oven-dried at 70°C for 72 h, cut to fit into beakers, submerged in 200 mL of Nanopure® deionized water, and refrigerated at 4°C for 72 h. While these methods do not reflect natural conditions, we felt that they were necessary to lessen the effects of several site-dependent confounding factors such as atmospheric deposition and stream water chemistry. For each leachate type (i.e., the same field site, species, and phenophase), 30 leaves were individually submerged in Nanopure® deionized water. To decrease costs, leachate samples of the same type were divided into groups of six with a total of 290 mL of leachate pooled evenly from the six samples. Thus, for each leachate type, there were five pooled samples to be analyzed for fluorescence. Pooled leachate samples (n = 75) were filtered into 40 mL amber glass vials using Whatman GF/F 47 mm 0.7 μm filters, which were placed into cold storage until fluorescence analysis.

2.3. Senescence Progression Ranking of Leachate Samples

Using the digital image of each leaf, percent greenness (G%) [*Richardson et al.*, 2007] was calculated using ImageJ based on the following equation:

$$G\% = \frac{\mathrm{GDN}}{\mathrm{RDN} + \mathrm{GDN} + \mathrm{BDN}} \times 100 \quad (1)$$

where GDN, RDN, and BDN are the green digital number (DN), red DN, and blue DN, respectively. Average G% was calculated for each pooled leachate sample. The plant senescence reflectance index (PSRI), which has been shown to be sensitive to senescence-induced reflectance changes [*Merzlyak et al.*, 1999], was calculated for all senescing and abscised leaves based on the wavelengths in the following equation:

$$\mathrm{PSRI} = \frac{\mathrm{Reflectance}_{680} - \mathrm{Reflectance}_{500}}{\mathrm{Reflectance}_{750}} \quad (2)$$

2.4. EEMs Preparation

Fluorescence and ultraviolet analysis of the leachate samples was conducted in the Watershed Sciences Research Group Laboratory at the University of Delaware. Samples were brought back up to room temperature, and 3-D-EEM scans were created on a Horiba Aqualog® Benchtop Fluorometer for DOM (Kyoto, Japan). EEMs were corrected using manufacturer-supplied correction factors and normalized by their respective Raman Areas. An inner-filtering correction was applied, and blanks were subtracted. For a more complete description of EEM preparation at this laboratory, see *Inamdar et al.* [2012]. Measured excitation wavelengths were from 239 to 551 nm with a frequency of 2 nm. Measured emission wavelengths were from approximately 250 to 828 nm with a frequency of approximately 4 nm.

2.5. PARAFAC Model Creation

PARAFAC modeling has been shown to be effective at locating common mathematically and chemically independent components in an EEMs data set, which represent either a single fluorophore or a group of highly related fluorophores [*Cory and McKnight*, 2005]. *Cory and McKnight* [2005] developed the approach for modeling a EEMs data set using a PARAFAC model, and their 13-component model is generally considered universal and applicable to a variety of fluorescent DOM (FDOM) EEM data sets. *Singh et al.* [2013], however, found that PARAFAC models that are site specific are useful for identifying more subtle differences in the DOM in the data set.

A PARAFAC model was created using drEEM software [*Murphy et al.*, 2013]. For all EEM samples, excitation wavelengths <260 nm, excitation wavelengths >400 nm, and emission wavelengths >550 nm were removed due to noise, and Raman and Rayleigh scatter was removed using the drEEM "smootheem" function. Four- through 14-component models were attempted for the data set. PARAFAC models are often subjective, and there are different methods used to validate them [*Murphy et al.*, 2013]. The most commonly used validation method is via split-half analysis where the model validates if different subsets of the EEMs data set produce statistically identical PARAFAC models. A six-component model (n = 68) was validated using the random split analysis after removing seven samples with high leverages (two MD senescing yellow poplar, one

NC abscised beech, one senescing NC beech, and three green MD yellow poplar). As suggested by *Murphy et al.* [2013], the style of validation used was $S_4C_6T_3$, where six different data set halves were assembled to produce three validation tests. The drEEM "splitanalysis" function was run five times, using a convergence criterion of 10^{-9}, and with the "nonnegativity" constraint. Using the same methods, split-half analysis validations were attempted based on leaf phenophases, geographic origin, and tree species, but all failed. A scree test was performed to further validate six components. PARAFAC component proportions and loadings for each EEM sample in the entire data set of 75 were calculated. While PARAFAC alone is not able to identify the molecular structure of the fluorescent components, some components can be identified as protein-like (e.g., tryptophan and tyrosine) or humic-like using comparisons to previously identified fluorophores. The OpenFluor network serves as an accessible venue for this purpose [*Murphy et al.*, 2014]. The OpenFluor network was used to match validated PARAFAC components with ones previously identified in the literature [*Murphy et al.*, 2014].

Nonrandom variation can remain within a data set after the number of validated components is removed [*Cuss et al.*, 2016]. Thus, residuals were calculated for each EEM sample to explore potential variation in the data set that could be explained by an additional component and were analyzed using Kohonen's self-organizing maps [*Cuss et al.*, 2016]. Hit histograms were created for the SOM of lowest mean quantization error (mqe) and topographical error (tge) using groupings of phenophases, geographic origins, and tree species. Statistical analysis was performed using JMP Pro 10's analysis of variance (ANOVA) test to investigate whether component proportions differed significantly between geographic origins, between species, and between phenophases.

2.6. SOM Creation

SOMs were created for PARAFAC model residuals and PARAFAC component proportions for the complete data set of leachate samples using the Fluor_SOmap package [*Cuss and Guéguen*, 2016] and SOM Toolbox 2.0 [*Alhoniemi et al.*, 2005]. Because compositional data are contingent on a constant-sum constraint [*Aitchison and Egozcue*, 2005], SOMs have been shown to be more appropriate for use on PARAFAC component proportions than other standard multivariate statistical analysis techniques, which can often lead to spurious correlations [*Cuss and Guéguen*, 2016]. For each data type, one linearly initialized map and 15 randomly initialized maps were created using 100 rough tuning iterations and 5000 fine tuning iterations, which have been shown to be adequate for the data set size [*Cuss et al.*, 2016]. Maps with the lowest mqe and tge were chosen for evaluation, because mqe and tge are related to the similarity of model vectors to sample vectors and the proper grouping of the samples based on their similarity to neighbors, respectively. The mqe and tge values for all created maps can be found in Table S2. Map dimensions and size for each randomly initiated map were based on those of the linearly initiated map created using the Fluor_SOmap package [*Cuss and Guéguen*, 2016]. Major features of the SOM (i.e., phenophase separation and species separation) were also present on other SOMs that had a relatively good fitting (tqe values of 0 and mqe values less than 0.833), indicating that the sample positions were relatively stable across the top SOMs. Hit histograms were created from groupings based on phenophase, geographic-specific population, and cluster. General patterns of senescence were identified on each SOM for each leaf origin based on G% and PSRI. For an in-depth tutorial on SOM creation, readers are referred to *Cuss and Guéguen* [2016]. We employed a step-by-step protocol to interpret our SOM results (Figure 1). Through the comparison of the component planes (step 3 in Figure 1), the numeric influence of each component on the fluorescence of a sample can be simultaneously visualized for all variable distributions. This is another major benefit of SOMs compared to many other multivariate approaches, such as principal component analysis.

3. Results

3.1. PARAFAC Model

We created a six-component PARAFAC model that validated via random split analysis and accounted for 99.55% of the variation (Figure 2). A model would not validate when splits were based on phenophase, species, or geographic origin. When the yellow poplar leaf litter leachate EEMs were removed from the data set, a five-component PARAFAC model validated with the same components as the first five in the full six-component model (for component spectra of the beech-only model see Figure S1 in the supporting information). The five-component, beech-only model also did not validate when splits were based on phenophase or geographic origin. Component 1 (C1) was found to match other components that were identified as tyrosine-

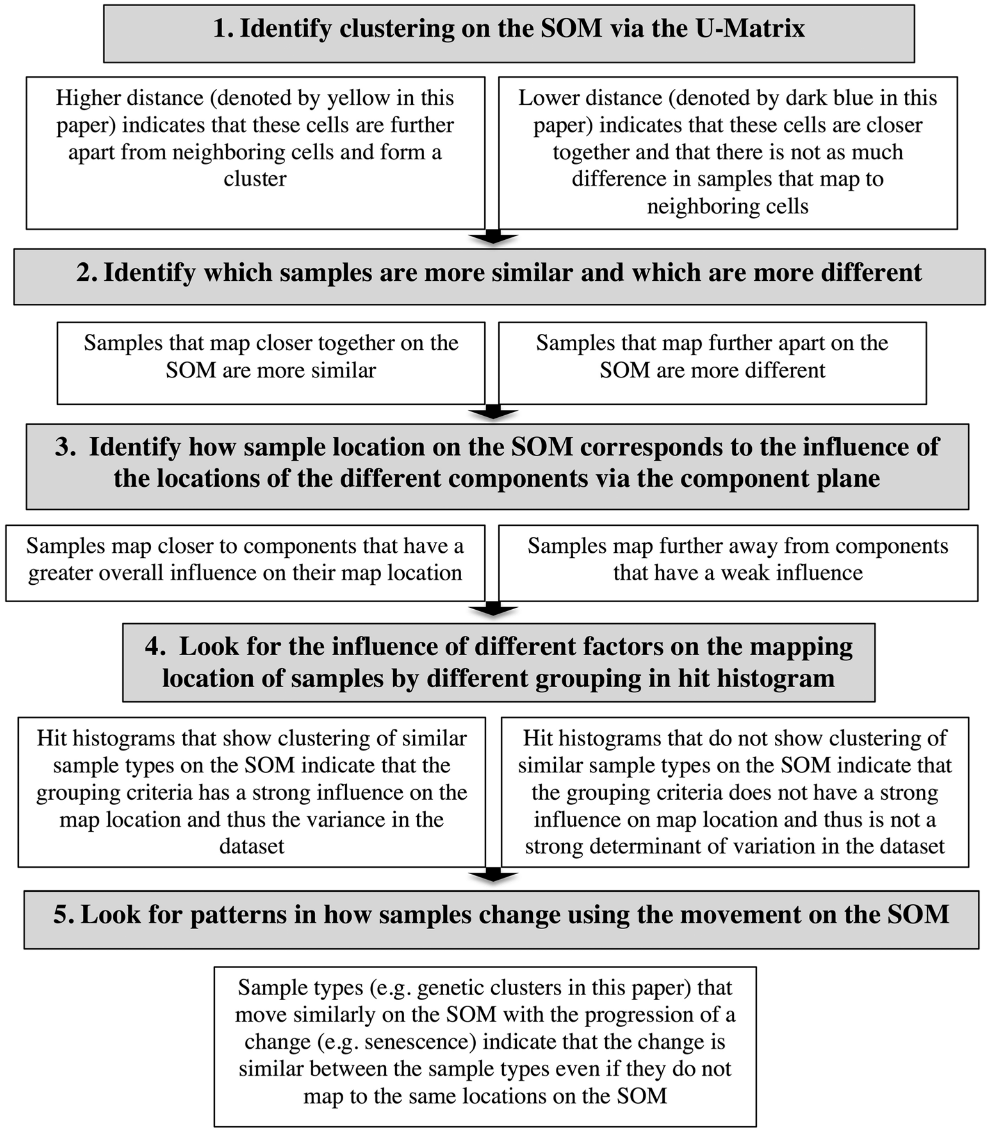

Figure 1. Summary of important steps for interpreting SOMs of multivariate data.

like and protein-like [*Murphy et al*., 2011; *Yamashita et al*., 2013]. Component 2 (C2) was the most ubiquitous of our components in other models and has been found to be tryptophan-like and protein-like [*Cawley et al*., 2012; *Lapierre and del Giorgio*, 2014]. Component 3 (C3) did not match any of the components in the published PARAFAC models on OpenFluor but was identified as protein-like [*Cory and McKnight*, 2005]. Component 4 (C4) matched a component of terrestrial origins [*Walker et al*., 2013] that was identified as humic-like [*Walker et al*., 2009]. Component 5 (C5) and component 6 (C6) were found to match humic-like components [*Yamashita et al*., 2013; *Kothawala et al*., 2012; *Lapierre and del Giorgio*, 2014]. On the SOM of the residuals, residuals are grouped by phenophase (Figure 3b), while there is little geographic-origin-dependent grouping of residuals (Figure 3c). Thus, variation in phenophase and not origin (geographic nor species) remains in the data set after six components are removed (Figure 3). Neuron residual EEMs corresponding to the corners and other representative neurons of the map (Figure 3) are shown in Figure 4 and show that nonrandom peak-like patterns of variance still remain in the residuals even after the six components have been applied.

The majority of the PARAFAC component proportions were significantly different between phenophases, with the exception of C6, and C4 and C6 when only MD leaf litter leachate EEMs were considered (Table 1). Within the beech-only leachate data set, none of the component proportions were significantly

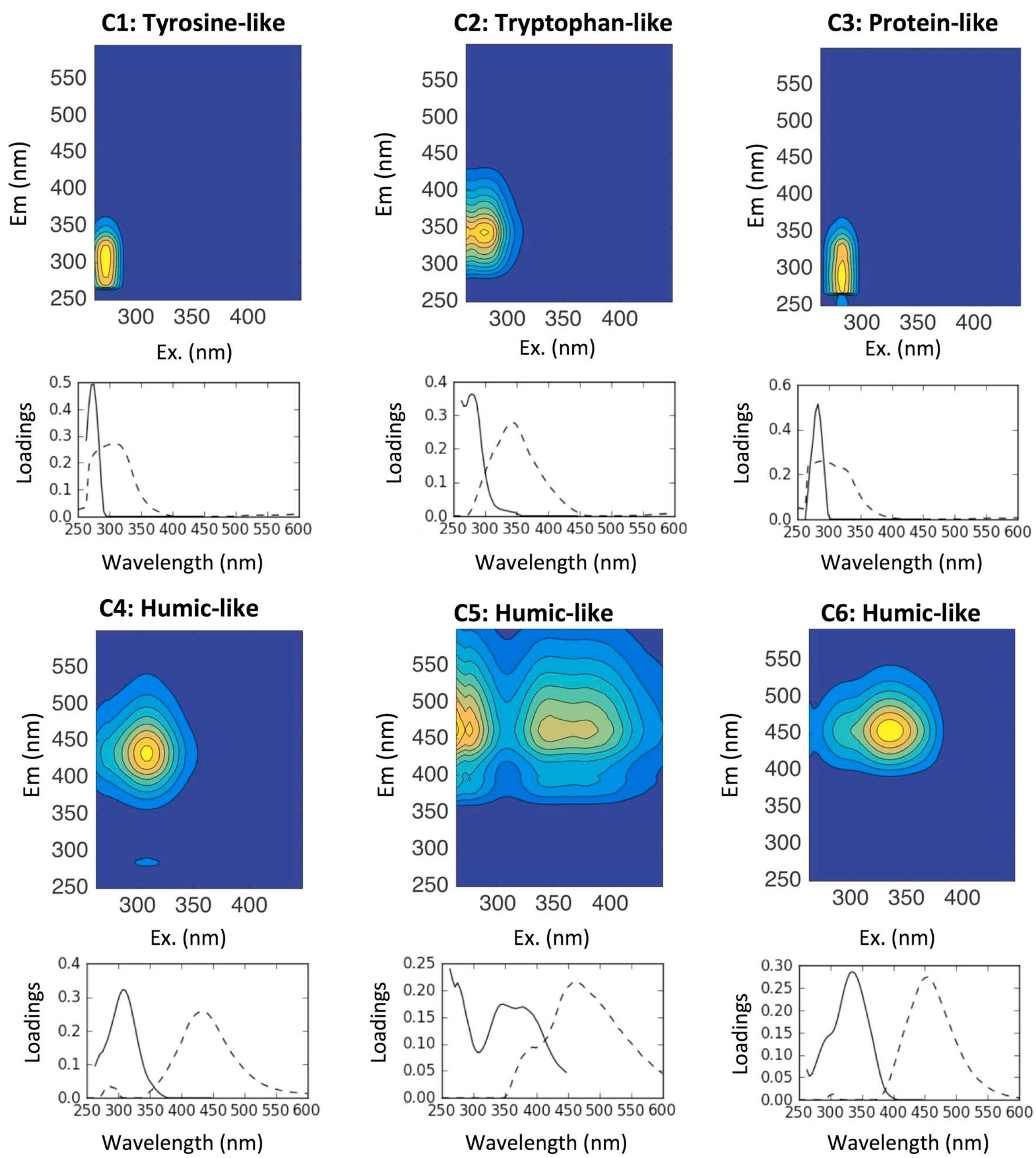

Figure 2. Parallel factor analysis (PARAFAC) components found in data set including both beech and yellow poplar.

related to the geographic origin. C1 and C5 were significantly different when the leachate EEMs were classified based on beech genetic cluster. Using only the MD leachate EEMs data set, C3, C4, and C6 were significantly different between beech and yellow poplar EEMs. The SOM of the component proportions was chosen with the lowest mqe of 0.8075 and the lowest tge of 0 (Figure 5a). Hit histograms were created that showed the distribution of leachate EEMs locations on the SOM based on phenophase (Figure 5c) and origin (Figure 5d). EEMs of the same phenophase map closer together on the SOM than apart (Figure 5c), but EEM location on the SOM is not divided by geographic origin (Figure 5d). Thus, leachate EEM location on the SOM was highly determined by phenophase and species but not by geographic origin.

3.2. Tracking Senescence

Measured MD beech leaves were found to have a G% range of 52.02% to 29.99% for green to freshly abscised leaves. RI beech leaves were found to have a G% range of 50.51% to 31.78%. VT beech leaves had a larger

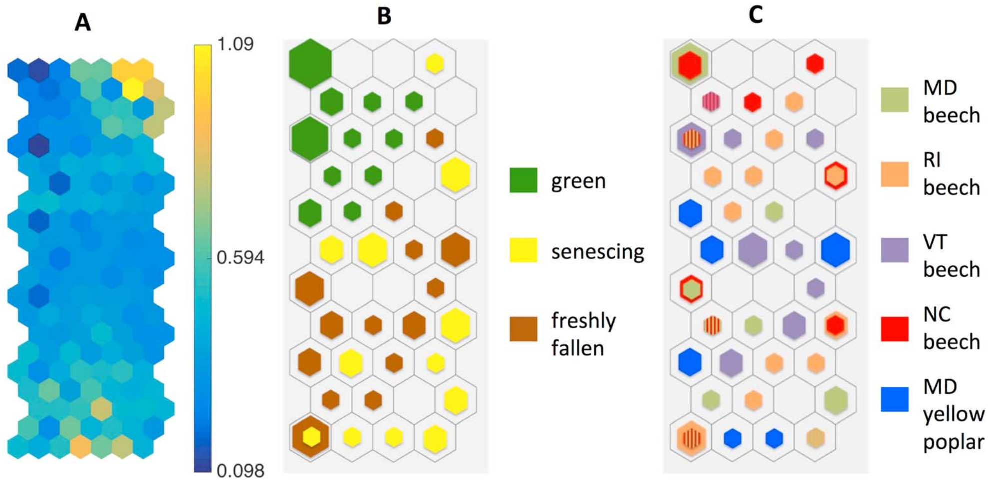

Figure 3. SOM of residuals for six-component PARAFAC model. (a) *U*-matrix where the color scale indicates distances between neighboring cells. (b) Hit histogram displaying the locations of residuals based on phenophase. (c) Hit histogram displaying the locations of residuals based on leaf origin. In the hit histograms, the size of the hexagon is proportional to the number of associated residual EEMs, ranging from one to six. The striped hexagons indicate that equal numbers of two geographic origins (of the hexagon size) mapped to that cell.

range of 64.08% to 30.41%. Similarly, NC beech leaves had a range of 62.35% to 29.34%. MD yellow poplar leaves had the largest range of 63.34% to 29.20%. Stacked area charts (Figure 6) and the SOM (Figure 7) illustrate how the component proportions change for each leachate EEM origin as the G% decreases and PSRI increases. The similar change in C1 in MD beech and RI beech is contrasted with the similar change in C1 in NC beech and VT beech (Figure 6). The proportions of C1, C2, C3, and C4 remain relatively equal to each other in MD yellow poplar for each phenophase, but the fluorescence is more dominated by C6 in senescing and abscised leaves (Figure 6).

When senescence progression is quantified by PSRI and G% (Figure 7), all green beech leaves start out in cells in the bottom left of the SOM (cells 7-1, 8-1, 8-2, and 8-3). At this location on the SOM, the component planes (Figure 5b) show that component 1 has the strongest influence on the fluorescence of green leaves. Component 1 accounts for approximately 50–55% of the total fluorescence composition of the green samples, C2 12–15%, C3 15–20%, C4 10–15%, C5 $<$ 1%, and C6 approximately 2% (Figure 5b, color bar legend). As senescence is quantified by PSRI, RI and MD beech (orange and sage colored arrows) both moved toward the top left 1-1 cell (row number 1–column number 1) on the SOM toward C2 and C3 and then moved more right toward C5 (Figure 7a). NC and VT beech (red and purple colored arrows) both moved initially toward cells 3-4 and 4-1, respectively (Figure 7a). All four beech geographic origins started at the bottom, moved upward on the SOM, and then downward, ending more toward the middle. When quantified by G% (Figure 7b), RI and MD beech (orange and sage colored arrows) both went to the top left cell 1-1, corresponding to C2 and C3, before eventually ending in cell 1-5, corresponding to C5. VT and NC beech (purple and red colored arrows) did not move as close to the top left corner (C2 and C3) as the RI and MD beech populations did and overall remained more central, before ending up toward the top right at C5 (Figure 7b). MD yellow poplar (blue colored arrow) had a similar pattern for senescence quantifications based on both PSRI and G% and started at the bottom-right in cell 8-5 and then moved upward finishing in cell 4-5, which corresponded to C6 (Figure 7b), staying toward the right of the SOM. Both the stacked area charts and the SOM showed similar behavior in the change in PARAFAC components with decreased G% throughout autumn, but the SOM allowed for better visualization of the patterns.

4. Discussion

4.1. Impacts of Changes in Phenology

Climate change is expected to increase the intensity and occurrence of numerous forest disturbances, including fire, herbivory, and severe weather events [*Dale et al.*, 2001]. Weather events that are associated with strong winds, such as hurricanes, windstorms, and thunderstorms, have the potential to result in greenfall

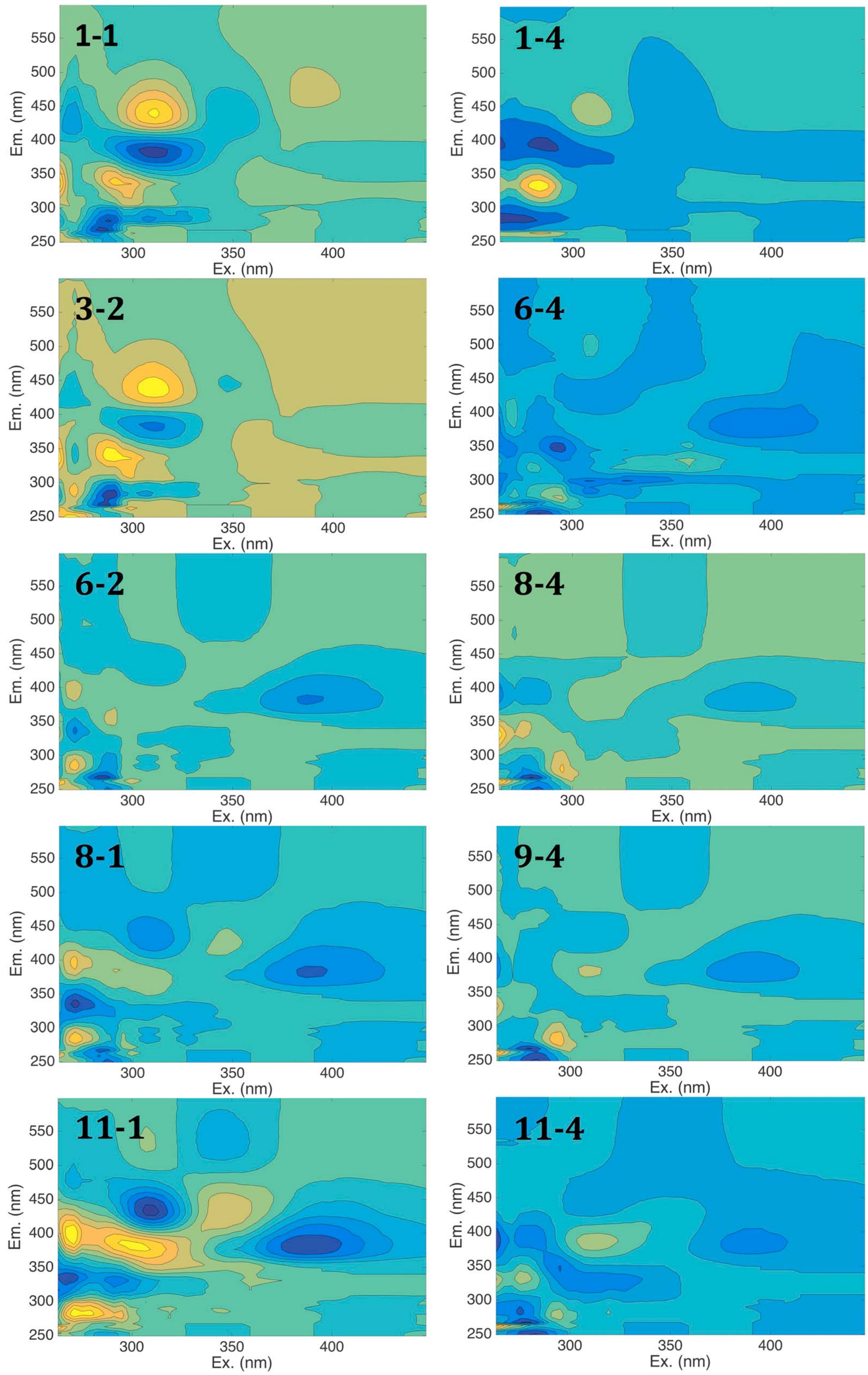

Figure 4. Representative neuron EEMs corresponding to varying parts of the SOM of PARAFAC model residuals. The identifying numbers refer to the row number and column number on the SOM with the top row on the SOM being row 1 and the left column being column 1.

Table 1. Results of ANOVA Tests on PARAFAC Component Proportions to Check for Significant ($p < 0.05$) Relationships[a]

		All			Maryland					
		Phenophase			Phenophase			Species		
	ID	d.f.	*F*	*p*	d.f.	*F*	*p*	d.f.	*F*	*p*
C1	Tyrosine-like	2	38.32	**<0.0001**	2	22.11	**<0.0001**	1	1.54	0.2243
C2	Tryptophan-like	2	25.52	**<0.0001**	2	10.52	**0.0004**	1	0.22	0.6391
C3	Protein-like	2	13.19	**<0.0001**	2	4.08	**0.0282**	1	26.78	**<0.0001**
C4	Humic-like	2	4.29	**0.0173**	2	2.43	0.1068	1	8.31	**0.0075**
C5	Humic-like	2	49.10	**<0.0001**	2	7.86	**0.0020**	1	3.43	0.0744
C6	Humic-like	2	1.9147	0.1548	2	1.65	0.2108	1	28.92	**<0.0001**
		Beech only								
		Phenophase			Geographic Origin			Genetic Cluster		
		d.f.	*F*	*p*	d.f.	*F*	*p*	d.f.	*F*	*p*
C1	Tyrosine-like	2	49.65	**<0.0001**	3	1.88	0.1431	1	4.36	**0.0411**
C2	Tryptophan-like	2	33.93	**<0.0001**	3	1.02	0.3891	1	2.94	0.0918
C3	Protein-like	2	15.92	**<0.0001**	3	2.00	0.1248	1	1.16	0.2866
C4	Humic-like	2	18.46	**<0.0001**	3	0.82	0.4889	1	0.14	0.7105
C5	Humic-like	2	83.29	**<0.0001**	3	2.01	0.1223	1	4.85	**0.0316**

[a]Significant *P* values are in bold. d.f. refers to degrees of freedom.

from trees. Additionally, delayed autumn senescence and leaf abscission will increase the risk of premature leaf abscission by autumn frosts [*Fracheboud et al*., 2009]. Therefore, it is important to understand how these disturbances will ultimately affect the nutrients that leach from fallen leaves and how the timing of the disturbances could affect biogeochemical cycling in forests. As autumn progresses, the leaf litter leachate samples move upward and eventually to the right on the SOM, further away from the tyrosine-like C1 in the bottom left on the SOM (Figure 5b). Abscised beech leaf litter leachates mapped more in the top-right quadrant of the SOM closer to C5, which is expected due to the humic-nature of C5. As the leaves senesce and breakdown, the degree of humification should increase. Our work has shown that the changing contributions of each PARAFAC component to the overall sample fluorescence throughout autumn are more apparent when mapped on the SOM.

In addition to changing the timing of phenological events, climate change is also expected to alter phenological processes, such as nutrient resorption [*Estiarte and Peñuelas*, 2015]. The resorption of limiting nutrients, such as N and phosphorus (P), in autumn is important because almost all of initial spring growth is based on the nutrients resorbed in the autumn before leaf abscission [*Neilsen et al*., 1997]. N and P that are not resorbed leach from abscised leaves, returning to the soil or nearby water bodies, affecting growth elsewhere. Specific enzymes facilitate the deamination of nucleotides and proteins to free ammonia, which is then used to synthesize amino acids [*Estiarte and Peñuelas*, 2015]. In the dormant, winter season, the N is stored mainly as storage proteins and amino acids in bark and young twigs [*Estiarte and Peñuelas*, 2015].

The process of N resorption is likely seen in the changing of the PARAFAC component proportions in our data set and in the patterns on the SOM. The patterns on the SOMs are able to help compare the behavior of the different protein-like PARAFAC components in the autumn. EEMs on the SOMs map near PARAFAC components that have the greatest impact on the sample's fluorescence. Thus, the movement away from C1 with the onset of senescence indicates a lessening of the contribution of the tyrosine-like component to the sample's fluorescence and an increase in the contribution of the other components it approaches on the SOM. Many of the senescent leaf litter leachates mapped to the upper left area of the SOM (Figure 5c), indicating that C2 and C3 had higher proportions than C1 in many senescent leaves. This suggests that the tyrosine-like component is the first component to decrease in proportion in the leaf litter leachate and that compounds that comprise the tryptophan-like C2 and the protein-like C3 do not decrease throughout the autumn. This is further apparent in the PARAFAC component loadings (see Figure S2). This unequal changing in the proportions of C1, C2, and C3 in the beech leachate is also apparent in Figure 6. The proportion of C1 decreases for all four geographic origins between green and senescent leaves, while proportions of C2 and C3 increase, more in some geographic origins than in others. Thus, the impacts of severe weather events on the

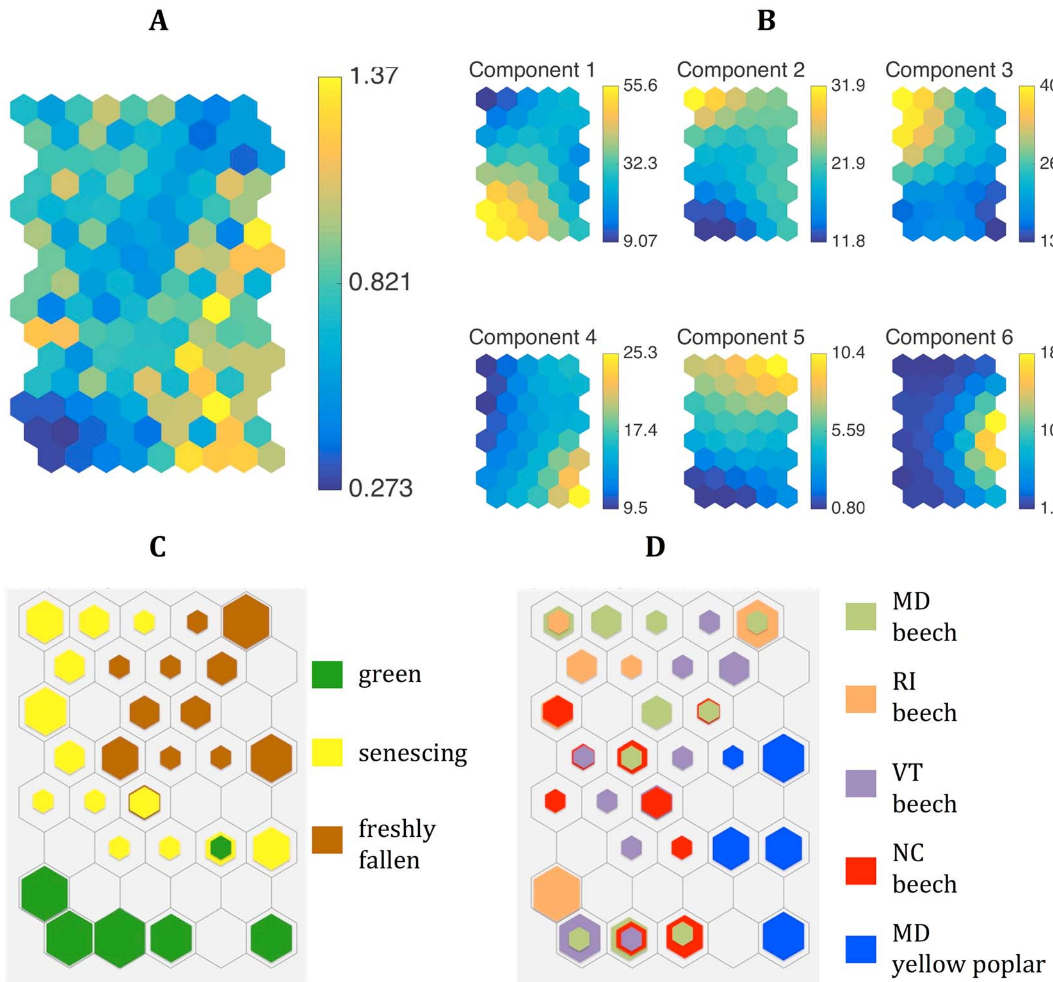

Figure 5. (a) SOM of PARAFAC component proportion *U*-matrix. (b) Component planes on the SOM. (c) Hit histogram displaying the locations of samples based on phenophase. (d) Hit histogram displaying the locations of samples based on leaf origin. In the hit histograms, the size of the hexagon is proportional to the number of associated samples, ranging from one to six. The color scale in Figure 5a indicates the distances between neighboring cells and in Figure 5b the percentage of total fluorescence composition for each component corresponding to the sample EEMs that are associated with the respective map unit in the fitted SOM.

biogeochemical cycling in a forest would differ depending on when the disturbance occurs. The FDOM in fallen green litter leachate would be predominately C1, senescing leaf litter leachate would have more fluorescence contribution from C2 and C3, and the naturally abscised leaves would have more humic-like fluorescence.

It is necessary to note the importance of identifying components as protein-like instead of simply as proteins and amino acids. Several compounds have been found to fluoresce with similar spectral signatures to proteins and amino acids [*Aiken*, 2014]. These additional compounds include tannins and lignin phenols [*Aiken*, 2014], which are characteristically present in leaves. Specifically, lignin phenols tend to share similar regions of the EEMs spectra with tryptophan and even more with tyrosine [*Hernes et al.*, 2009]. Therefore, while useful information regarding the behavior of some nitrogen-containing compounds is available, it is limited and not definitive. C1 is tyrosine-like, and thus, it was expected that as trees resorb nitrogen, which comprises amino acids like tyrosine [*Millard and Proe*, 1991], the tyrosine in the leachate should decrease, which is seen as the leachate sample groups transition away from the lower left on the SOM. Therefore, even though there are likely other nonprotein compounds contributing to fluorescence of C1, the expected behavior of tyrosine is still conserved and apparent. The concentrations of other nitrogen containing proteins and amino acids should also decrease with nitrogen resorption and senescence, but the proportions of C2 and C3 do not decrease as drastically as C1 did and even increase in fluorescence proportion in senescing leaves. This

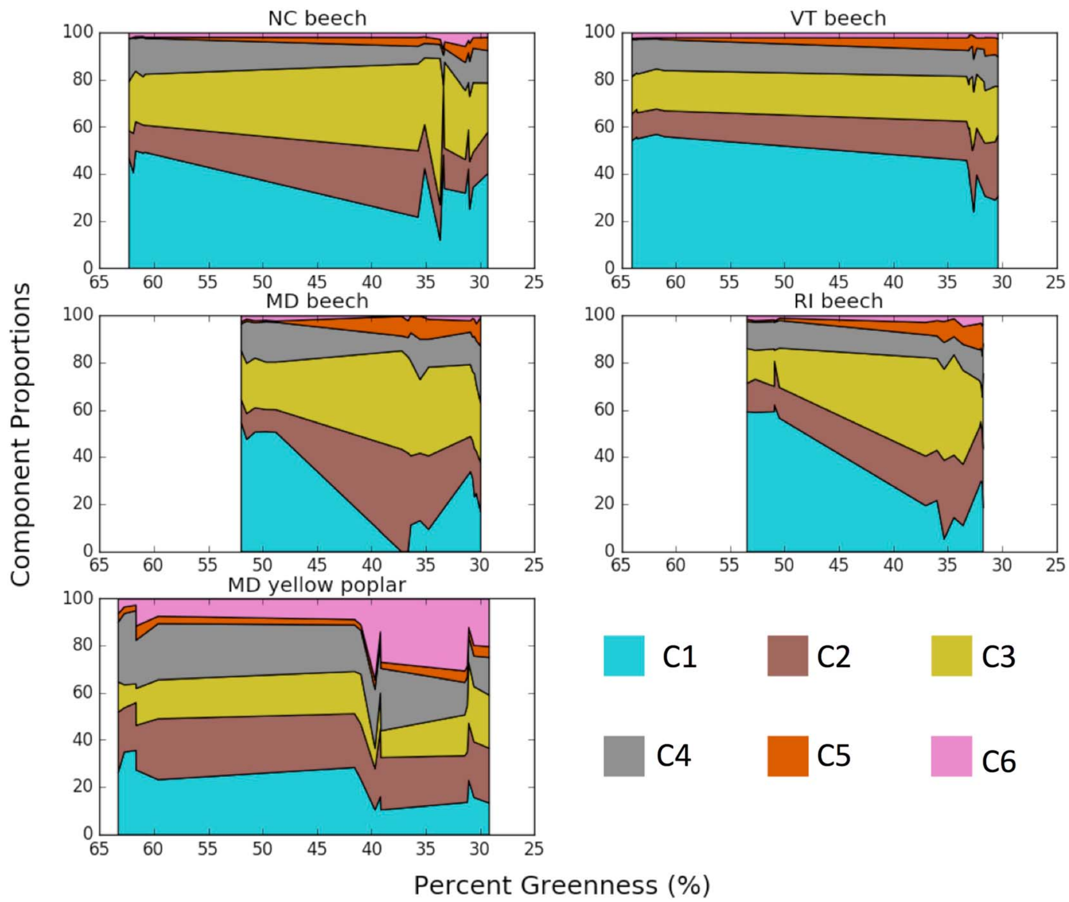

Figure 6. PARAFAC component proportions of the submerged leaf litter leachate samples of different state origins for varying averaged percent greenness.

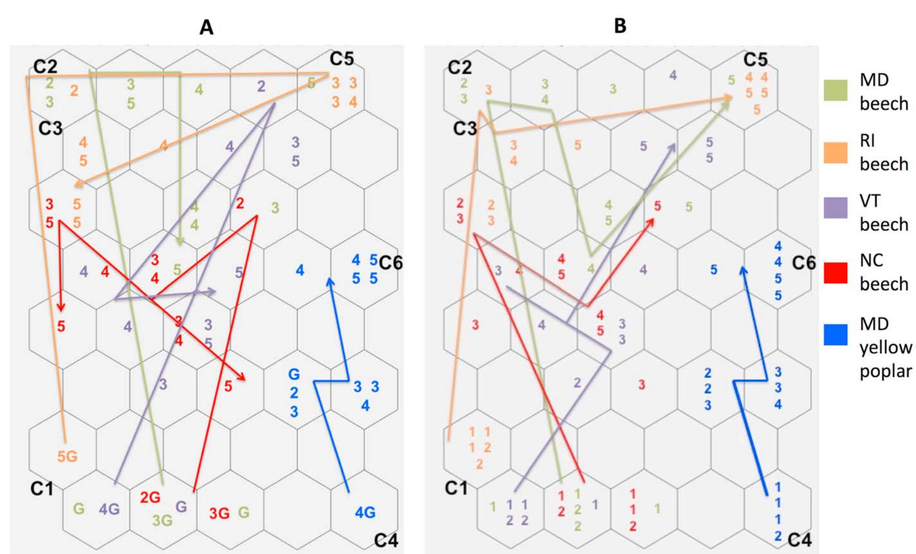

Figure 7. (a) SOM showing tracks in senescence for the different leaf origins based on changing PSRI. G representing green leaves. (b) SOM showing tracks in senescence based on changing G%. Samples were divided into five groups by order of senescence (based on the respective PSRI and G% values), with 5 being furthest in senescence progression. The C1–C6 labels indicate where the PARAFAC model components map on the SOM as determined by the component planes in Figure 5b.

either indicates that resorption of tryptophan and the proteins that contribute to C3 are not as strongly resorbed or that other nonprotein compounds are heavily affecting the fluorescence in the spectral regions of C2 and C3. Tyrosine and tryptophan might be broken down and resorbed differently, possibly from differences in their respective functions within leaves, different molecular weights, different biochemical limitations, and differences in structure. Either way, the climate change effects on the resorption of tyrosine would likely manifest in the PARAFAC analysis, while any changes to the resorption of tryptophan or the proteins possibly present in C3 would likely not be able to be seen in a PARAFAC model.

4.2. Impacts of Changes in Tree Community Composition

With changing local climates, tree species are more likely to migrate or face extinction than to adapt to the new climate niches [*Peterson et al*., 1999]. *Iverson and Prasad* [2002] modeled the future distribution of 76 common tree species in the eastern United States and found that the optimal ranges of the majority of tree species will move northward, which would cause different tree species to be present in an area. *Iverson and Prasad* [2002] found the potential area of habitat to decrease by 85.7% for beech trees and to slightly increase by 3.9% for yellow poplars by the year 2100. They found that beech trees could be almost eradicated throughout the United States, except for the southern Appalachian Mountains, whereas the range of yellow poplar could move northward 20–240 km depending on the climate model used. Thus, forests that currently contain beech trees could contain more yellow poplar in the future. Because C6 was only found in measurable amounts in yellow poplar leaf litter leachate, restructuring forest composition would increase the amount of C6 in nearby water bodies, especially with maturely abscised leaves. Similarly, C4 was significantly higher in yellow poplar leaf litter leachates than in beech leachates and could be expected to increase in the ecosystem with tree community composition restructuring. Oppositely, C3 was significantly higher in beech leachates, especially in the senescing leaves, than in yellow poplar leaves and would likely decrease with community composition restructuring. No apparent interspecific difference occurred in the SOM of the residuals, confirming that the six-component model likely accounts for any measurable interspecific variation. The significant interspecific differences for the three component proportions caused all yellow poplar samples to map to separate cells than beech on the SOM (Figure 5d). The placement of yellow poplar samples on the SOM transitions from cell 8-5 to cells 6-4 and 6-5 and then to cells 4-4 and 4-5 as the leaves changed from green to senescing to abscised. This indicates a shift from higher proportions of C4 and C1 toward higher proportions of C6. Thus, even though there are significant differences between C1 of the two genetic clusters of beech, the interspecific differences in the overall leachate FDOM are much more pronounced. The SOM method does an excellent job at illustrating the stark differences in the overall PARAFAC component proportions between beech and yellow poplar leaf litter leachates. Replacing beech trees with yellow poplar trees would alter the FDOM that enters water bodies via leaf litter leachate, especially the C3, C4, and C6 components.

Genetic variation in tree species, especially between distinct genetic clusters, has been shown to be primarily due to adaption to varying climates and not genetic drift [*Hall et al*., 2007]. *Kitamura and Kawano* [2001] determined that two distinct genetic clusters of American beech exist in the eastern United States. American beech trees in harsher environments typically regenerate asexually by root suckers, while mild-environment trees regenerate by sexual reproduction [*Held*, 1983; *Kitamura et al*., 2000]. Beech populations living in areas of former glaciation and in the high elevations of the Appalachian Mountains were found to be part of the harsher-environment genetic cluster, and those found elsewhere, specifically in the Gulf-coastal plain, eastern coastal plain, Piedmont Plateau and Ozark Plateau, were part of the mild-environment genetic cluster [*Kitamura and Kawano*, 2001]. Specifically *Kitamura et al*. [2001] found that beech populations growing in the Blue Ridge Mountains in Virginia and in the higher elevations of the Great Smoky Mountains in NC reproduced via root suckers and were genetically similar to populations growing in harsher northern environments. Our NC sample site is located in northwestern NC in the Blue Ridge Mountains and is at a higher elevation than *Kitamura et al*.'s [2001] sampled Great Smoky Mountain site of 1300 m elevation. *Kitamura and Kawano* [2001] specifically studied a beech population from VT and found studied individuals to be genetically adapted to harsher environments. Thus, in our analysis we characterized our NC and VT beech leaves as being from the harsh-environment genetic cluster. *Kitamura et al*. [2003] specifically studied beech populations within Maryland and found that they were all part of the mild-environment cluster. Although none of the Kitamura et al. papers examined beech populations in RI, it was assumed that our sampled RI population was from the mild-environment genetic cluster because it was well south of the latitude-based division between the harsh-environment and mild-environment clusters. For a map of the divide, readers are directed to *Kitamura and Kawano* [2001].

The difference between the two genetic clusters was especially evident in our sampled green leaves. Sampled green leaves from trees living in the harsher environments had much higher G% values than those from trees living in the more mild environments. The green leaves from the harsher environment had a mean G% of 62.21% (standard deviation of 2.503), and the green leaves from the milder environment had a mean G5 of 51.15% (standard deviation of 2.749), which was significantly lower with a P value of <0.001. All green leaves were collected before senescence began during the growing season so differences in G% were not due to early-onset senescence. There were also not visible signs of chlorosis or other symptoms that could affect the G%. The G% has been linked to photosynthetic capacity [*Richardson et al.*, 2009; *Sonnentag et al.*, 2012] due to varying amounts of the pigment chlorophyll, which is necessary for photosynthesis [*Lee et al.*, 2003]. Trees living in harsher environments have been known to adapt and acclimate, for example some trees compensate for a shorter growing season by having higher rates of photosynthesis per unit of leaf area [*Pallardy*, 2008]. Some of this is due to a tree's ability to acclimate to certain environmental stresses but has also been shown to be more related to advantageous genetic adaptations. *Ledig and Korbobo* [1983] found that the photosynthetic capacities from sugar maple progenies from high latitudes were higher than those from low latitudes even when all trees were grown in a uniform environment. This bolsters our hypothesis that genetic differences between the two genetic clusters result in differences in plant physiology. The use of PhenoCams [*Richardson et al.*, 2009] and other imagery sources could potentially be used to monitor the switch between genetic clusters present in a forest resulting from climate change due to the differences in G% in their green leaves.

In the ANOVA results, only C1 and C5 were found to differ significantly between the two beech genetic clusters. Even though only two of the five beech components were significantly different, C1 accounts for the most variation in the data set and contributes highest to the fluorescence in the green leaf litter leachate for all field sites. The significant difference in C1 between the two clusters is evident in Figure 6; all geographic origins start out with similar C1 proportions, but MD and RI C1 proportions quickly decrease to minimal amounts in senescing leaves, before slightly rising again in abscised leaves. In contrast, the proportion of C1 remains much more constant in NC and VT samples as G% decreases (Figure 6). The sharp decrease in C1 in MD and RI causes C1 and C2 to have greater influence on the FDOM in the sample, resulting in senescent MD and RI samples mapping to the top left corner of the SOM (Figures 5c and 5d). This is more evident when PSRI and G% are used to divide the samples from each field site into five levels of senescence. Quantifying senescence based on G% did a better job of showing patterns in senescence than when using PSRI (Figure 7). This was likely due to G% being based on the whole leaf whereas PSRI is based on reflectance at one part of the leaf where the portable spectrophotometer's leaf clip attached. Parts of a leaf senesce at different times and rates. Thus, the G% is a holistic measurement of senescence of the leaf and therefore more accurately shows the rough patterns of senescence for the different origins. We recommend that future studies that use spectral metrics obtained from a similar instrument to measure the spectral reflectances at multiple areas of a leaf to attempt to alleviate this issue. All green beech leachates begin at the bottom left of the SOM near C1 (Figure 7).

There is likely a genetic-cluster-dependent difference in resorption of tyrosine-based N, which causes the difference in behavior of C1. This difference is also seen in the PARAFAC component loadings in Figure S2. Phylogeny has been found to have a strong influence on resorption. Resorption efficiency in a species, or genetic cluster, is thought to be partially dependent on the ancestor's evolutionary selection for higher or lower resorption efficiencies [*Killingbeck*, 1996]. Trees with well-developed nutrient conservation and acquisition characteristics usually have decreased potential resorption and vice versa [*Killingbeck*, 2004]. When grown in common garden conditions, trees with colder native climates have been shown to have higher leaf N and P content than trees of the same species that originate from warmer climates, indicating an adaptation to native climates [*Chapin and Oechel*, 1983; *Reich et al.*, 1996; *Oleksyn et al.*, 1998]. Thus, it is likely that trees in VT and in high elevations of the Appalachian Mountains of NC have adapted to better conserve and acquire nutrients, which affects potential resorption. Additionally, the timing of nutrient resorption has an impact on the resorption efficiency: early abscission has been found to result in low resorption and late abscission in elevated resorption [*Killingbeck*, 2004]. The harsher environments of the NC and VT beech trees cause them to start to resorb and senesce earlier than MD and RI trees. *Bradshaw and Stettler* [1995] found that in *Populus* spp. 98% of total phenotypic variation in phenology was determined by genetic factors, while only 2% was due to environmental factors. Latitude-based differences in phenology remain even when trees are planted in common gardens [*Hall et al.*, 2007]. Thus, even though some of the differences in the apparent tyrosine resorption could be due to environmental factors, most is likely due to genetic differences. Another significant determinant of resorption is soil fertility, particularly total N, although the effect of high and low values on nitrogen resorption efficiency, especially in the short term, remains controversial in the literature [*Killingbeck*, 2004]. The NC and VT sample sites have similar total N, but RI and MD are substantially different from each other (for additional soil nutrient characteristics see Table S1). RI has the highest total N, and MD is almost 10 times lower. Thus, the concentration of total N in the soil does not appear to be an appreciable cause of the apparent genetic-cluster-based difference in tyrosine resorption.

It has been suggested that one of the options to mitigate the impacts of climate change on forests is to identify which genetic clusters within a species are most adapted to the predicted new local climates and replace trees in a forest with other trees of the same species but of the more optimal genetic cluster [*O'Neill et al.*, 2008]. Especially for tree species such as beech that are expected to face near eradication in potential climate change scenarios, this artificial selection of trees could help maintain species ranges. *McLachlan et al.* [2005] found that past migration rates of beech trees were substantially slower than those required to conform to the rate of current anthropogenic-caused climate change. Our results indicate that while not as severe as substituting species, switching genetic clusters would likely affect the biogeochemical cycling in forests, especially the behavior of the tyrosine-like C1 component throughout autumn. This could likely be preferable to the potential eradication of the beech species in most of the United States, but altered forests would need to be monitored for changes in nutrient cycling.

In contrast to the other comparisons, none of the components significantly differed between geographic origins for the beech. There was also no geographic-origin-dependent variation remaining in the data set as evidenced by apparent randomness of geographic origin locations on the SOM of PARAFAC model residuals (Figure 3c). There is also little apparent order to the mapping of beech samples by geographic origin on the SOM of PARAFAC component proportions (Figure 5d), especially when compared to differentiation by phenophase. Although there are no distinct clusters of beech samples by geographic origin on the SOM, there are some geographic-dependent differences evident on the SOM. This is more apparent when comparing the position on the component proportion SOM (Figure 6). Mainly, the C1 proportion in the VT leaves decreases the least amount among the geographic origins and VT's C2 and C3 proportions barely increase with time. This explains why no VT samples mapped close to C2 and C3 on the SOM (Figure 5d). Similarly, the proportions of C3 did not change as much for NC as G% decreased, which explains why no NC samples mapped to the top-left part of the SOM (cells 1-1, 1-2, and 1-3). The proportion of C3 in NC did slightly increase between green leaves and senesced leaves, and thus, NC samples shifted closer to the top-left of the SOM than VT leaves. Therefore, while nutrient cycling models should try to account for differences between phenophase, species, and genetic clusters, they can be simplified by not including variation between sites that are in relatively similar ecosystem types.

5. Conclusions

Based on an EEMs data set of leaf litter leachate from two species, four geographic origins, and three phenophases, a six-component PARAFAC model was validated. The model was validated by random split analysis, but not when the splits were based on phenophase or geographic origin, indicating that the component proportions change throughout the fall and differ somewhat between field sites. Some phenophase-dependent variation remained in the data set after six components were fitted, but little geographic- or species-dependent variation remained. Overall, samples transitioned from more protein-like components to more humic-like components with senescence and abscission. The proportions of most components differed significantly between phenophases. In contrast, none of the component proportions differed significantly between field sites. This likely indicates that there were no substantial variations in factors such as soil or atmospheric deposition between field sites to influence FDOM in the leachate. C1, which identified as tyrosine-like, did differ significantly ($p = 0.04$) between beech genetic clusters. This difference in tyrosine resorption is likely due to both genetic differences and environmental factors such as the timing of resorption and abscission. A genetic shift in the local population from trees adapted to harsh environments to trees adapted to mild environments could result in less tyrosine lost in abscised leaf litter. Even though there were some differences between the beech genetic clusters, the differences were not as pronounced as those between beech and yellow poplar and between phenophases. The measured leaf litter leachates were created in an artificial laboratory environment that does not necessarily reflect the behavior of the components in situ, and additional work should be attempted to see if similar seasonal patterns are evident in forested streams. However, the statistically significant differences between leachates from the three phenophases found under laboratory conditions foreshadow how dynamic the contributions of DOM from intraspecific and interspecific sources might be under natural circumstances. The effect of the variability of this input on primary production and nutrient cycling should not be overlooked.

A warming climate is expected to result in changes in tree phenology [*Estiarte and Peñuelas*, 2015] and in the distribution range of tree species [*Iverson and Prasad*, 2002]. This study shows that phenology and species have a greater influence on the FDOM, when characterized by PARAFAC components, than genetic clusters within a species. However, there are some fluorophores, such as the tyrosine-like component shown here, in leaf litter leachate that would likely be altered with only changing the genetic identity of members of a forest without changing the species composition. These potential intraspecific differences should be taken into consideration when evaluating the affects of climate change on watersheds and ecosystems. SOMs served as a useful technique to visualize these differences and should be utilized more in future EEMs studies, especially those utilizing PARAFAC components. Leaf litter leachate plays a major role in the biogeochemical and hydrologic cycles in forested watersheds and changes in FDOM can have major impacts on ecosystem processes. This study is, however, only based on four field sites, and we encourage further investigation into how shifting intraspecific and interspecific genetic cluster ranges will affect the FDOM in the end-member leaf litter leachate, especially in the fall where FDOM was shown to change considerably, and how resultant changes in fluorophores influence other ecosystem processes.

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